

ACADEMIC QUALIFICATIONS			
Year	Degree/Certificate	Institute	CPI/%
2025	B.Tech	Indian Institute of Technology Kanpur	6.1/10
2020	Class XII (MSBSHSE)	Dr. Ambedkar College, Nagpur	81.38%
2018	Class X (CBSE)	Bhavan's Bhagwandas Purohit Vidya Mandir, Nagpur	92.8%
WORK EXPERIENCE			
AI/ML Research Intern Airawat Research Foundation MOHUA			(June'25-Jul'25)
Objective	● Propose AI/ML solutions to key challenges in the Housing for All (HFA) division of MOHUA using real urban data		
Initiative	● To classify 1 Cr+ urban households as Kuccha/Pucca using unsupervised learning through satellite imagery ● Built OCR + NLP pipeline for auto-validation of income, land, and loan documents with integrated PDF compression ● To develop image-matching tool to verify beneficiaries by matching Aadhaar photos with geo-tagged images		
Outcome	● Enabled targeted PMAY-U outreach through vulnerability mapping of urban zones ● Validated 12K+ documents, cutting manual errors and speeding up verification ● Enhanced fraud detection via automated beneficiary identity checks		
KEY PROJECTS			
MBA737A Course Project Mentor: Dr. Abhinava Tripathi 			(Aug'23 - Nov'23)
Training index model for security price prediction Time Series Forecasting with R			
● Performed a time series analysis on ABC Firm and nifty index prices, modeling log returns and volatility trends using R ● Quantified statistical deviations (skewness : 1.5 vs. 0.8, kurtosis : 4.2 vs. 3.1) between security and market benchmarks ● Trained MLR , SLR , SLR2 , and Naive models; selected MLR for superior correlation (R=0.85) and forecasting accuracy ● Validated MLR robustness (VIF<2 , adjusted R²=0.82) & significance ($\beta = 1.2, p < 0.01$) to ensure reliable price predictions ● Mitigated multicollinearity , heteroscedasticity and autocorrelation (Durbin-Watson DW=1.98) for model stability			
CS771A Course Project Mentor: Prof. Purushottam Kar 			(Jan'25 - Apr'25)
Companion Arbiter PUF ML & Cloud Optimization			
● Derived a mathematical model to break a CAR-PUF using a single linear model with a feature vector dimension of 528 ● Deployed logistic regression model to analyse 40,000 training CRPs and 10,000 test CRPs , achieving 99.31% accuracy ● Optimized the mapping function's time complexity, reducing training time by 30% and mapping time by 37% ● Reported outcomes of experiments with LinearSVC & LogisticRegression , analyzing loss hyperparameters & C values			
Detecting Credit Card fraud Feature Engineering, Imbalanced Data Handling 			(Dec'23 - Apr'24)
● Conducted market research and Porter's 5 Forces analysis for market entry in the financial fraud detection industry ● Engineered features from synthetic data based on criminals' limitations and money laundering patterns ● Handled class imbalance by SMOTE , deployed decision tree algorithm to achieve F1 Score of 0.81 and 0.74 respectively			
MSE643A Course Project Mentor: Dr. Krishanu Biswas 			(Jan'25 - Apr'25)
ML for Perovskite Structure Data Modeling & Feature Selection			
● Built Random Forest & XGBoost models to classify 4 perovskite structures, achieving 82.6% accuracy & 0.82 F1-score ● Performed data preprocessing of 5000+ samples with feature engineering (ionic radii, bond lengths, etc.) & label encoding ● Analyzed feature importance to identify Goldschmidt tolerance factor as the key predictor driving crystal structure classification			
TECHNICAL SKILLS			
● Programming Languages: C, Python, SQL, MATLAB, R ● Tools & Platforms: Power BI, Excel, Tableau, AWS (Basics), Git ● Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Tensorflow, PyTorch, SciPy ● Techniques: Data Modeling, Feature Engineering, Time Series Forecasting, Fraud Detection ● Coursework: Databases & SQL, Machine Learning, Google Data Analytics, Advanced Learning Algorithms, Linear Algebra & Ordinary Differential Equations, Partial Differential Equation, Machine Learning Applications in Financial Markets			
POSITION OF RESPONSIBILTY			
Organizer, Events & Competitions Techkriti'24 Asia's Largest Technical & Entrepreneurial Festival			(Nov'23 - Apr'24)
● Handled more than 1000 participants during the conduction of Techkriti, one of Asia's most prominent technical festivals ● Worked as a part of the managing unit of various events, competitions, workshops, and performances during the festival ● Assisted heads in generating a revenue of INR 20 lakhs by managing an online registration portal for the participants ● Organized, The National Aeromodelling Workshop in association with BOEING comprising 150+ participants			
Organizer, Events & Competitions Udghosh'23 Asia's Largest Sports Festival			(Aug'23 - Nov'23)
● Leading a team of 30+ undergraduates to ensure smooth conduction of formal and informal events of the sports fest ● Managed the data and prepared schedules for every team competing in the competition in a number of sports ● Witnessed remarkable increase in reach and unprecedented participants from 450+ colleges with 100K+ footfalls			
EXTRA-CURRICULAR ACTIVITIES			
Secretary Esports society, IIT Kanpur			(Dec'22 - Apr'23)
● Member of the E-sports Club, focused on providing gamers with career opportunities and professional growth ● Managed the INTER-IIT Valorous Tournament valuation, which featured over 500 participants from 22 different IITs ● Organized a talk session featuring a Sikh Warrior, supported by sponsorships over INR 100,000 from Red Bull and Winzo			